SEGi COLLEGE
SUBANG JAYA

DIPLOMA IN INFORMATION TECHNOLOGY

APRIL 2026
SKILL-BASED EXAMINATION

PROGRAMMING FUNDAMENTAL

SUBJECT CODE: DIT1124

TIME ALLOWED: 3 HOURS

DATE: 6 APRIL 2026 **TIME: 9:30AM-11:30AM**

INSTRUCTIONS TO CANDIDATE:

1. Answer **ALL** questions.
2. ALL answers must be written on the question paper and submitted.
3. Dictionaries, Crying and Tantrum are **NOT ALLOWED**.

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TOTAL MARKS

SECTION

A :

B :

C :

D :

E :

SECTION A – MCQ **(TOTAL: 10 MARKS)**

Questions	Answer
1.	**B**
2.	**A**
3.	**C**
4.	**B**
5.	**A**
6.	**A**
7.	**D**
8.	**C**
9.	**D**
10.	**D**

SECTION B – True/False and Explanation

(TOTAL: 10 MARKS)

No	Answer	Explanation
1.	TRUE	Variable must be declared before starting a code in C++
2.	FALSE	"= =" Is used to compare two value
3.	FALSE	"for" loop is repeated if the condition is true
4.	TRUE	Array can store any data type that we assign in the code
5.	TRUE	Flowchart helps programmers to visualize the program flow and how it works. It's like a blueprint for a code.

SECTION C – Tracing Errors with Explanation **(TOTAL: 20 MARKS)**

QUESTION 1

```
1   #include <iostream>
2   using namespace std;
3
4   int main() {
5         int marks[3];
6         int total;
7         float avg;
8
9         cout << "Enter 3 marks:"<< endl;
10        for (int i = 1; i < 3; i++) {
11              cin >> marks[i]:
12              total = total + marks [i];
13        }
14
15        avg = total / 3;
16
17        cout << endl <<"Final Total: " << total << endl;
18        cout << Average Mark: " << avg << endl;
19
20        if (avg = 50)
21              cout << "Pass";
22        else
23              cout << "Fail";
24
25        return 0;
26  }
```

1.1 Based on above code, what is the meaning of "**endl**"? **(1 marks)**

Answer: "endl" means endline, it is use to end an argument in the code before starting a new argument in the next line of code.

1.2 Based on above code, identify the relevant errors, explain them and rewrite the corrected codes in table as shown below. **(10 marks)**

Answer:

Line	Explanation	Corrected Code

9	Cout is above "for" loop. Need to be below the "for" loop.	`For ( int i = 1; i <= 3; i++) {`
		`cout <<"Enter 3 marks:" << endl;`
		`}`

20	No curly bracket	`if (avg = 50) {`
		`  cout<< "Pass";`
		`}`
		`Else{`
		`  cout<<"fail";`
		`}`

1.3 Based on answers in 1.2, what is the purpose of this code? **(4 marks)**

Answer: the purpose of this code is to calculate the average grade of 3 marks. Looping the marks 3 time to get the user to input the marks and then the program will calculate "avg = total / 3 ", total marks divide by 3 to get the average marks.

1.4 Based on the corrected code, what is the final output if the input marks are: 50, 90, 35? You are required to use Tracing Table to support your answer. **(5 marks)**

Answer:

LINE	Marks[i]	i < 3	Total	Avg = total / 3
1	**50**	**True**	**50**	
2	**90**	**True**	**140**	
3	**35**	**true**	**175**	
4				**58.3**

The final output is 58.3

(Total: 20 marks)

SECTION D – Tracing Output **(TOTAL: 20 MARKS)**

QUESTION 1

```
    //Code nested for Loop

1   #include <iostream>
2   using namespace std;
3
4   int main() {
5         int total = 0;
6
7         for (int i = 0; i <= 5; i++)
8         {
9             if (i > 2)
10            {
11                total += 2;
12             }
13            else
14            {
15                 total += i;
16          }
17        }
18        cout << "Total: " << total;
19
20      return 0;
21  }
```

2.1 Based on above segment code, create tracing table to show the value of each variable during iteration of the loop. **(16 marks)**

Answer:

LINE	MAIN		SUB		TOTAL
	Condition	i > 2 +=2	Condition	i < 5 +=i	
1	F		T	0	1
2	F		T	1	2
3	F		T	2	3
4	T	3	T	3	6
5					

2.2 Based on your answer in 2.1, provide the final values for "i". **(1 marks)**

Answer: 3

2.3 Based on your answer in 2.1, provide the final value for "total". **(1 marks)**

Answer: 6

2.4 Explain what happens to the variable total when the following statements are executed inside a loop:

(2 marks)

- "`total += 2`"
- "`total += i`"

Answer:
"total += 2" is to add value of 2 to the int i

"total += i" is to add value of i to the int i.

(Total: 40 marks)

SECTION E – Case Study **(TOTAL: 40 MARKS)**

You are assigned to develop a C++ program that allows users to enter an integer number, check the number either positive or negative. If the integer number is positive, determine whether it is odd or even. Then display relevant messages such as "Positive number is an Even Number", "Positive Number is an Odd Number", and "The number is a Negative Number!". The requirements are as follow:

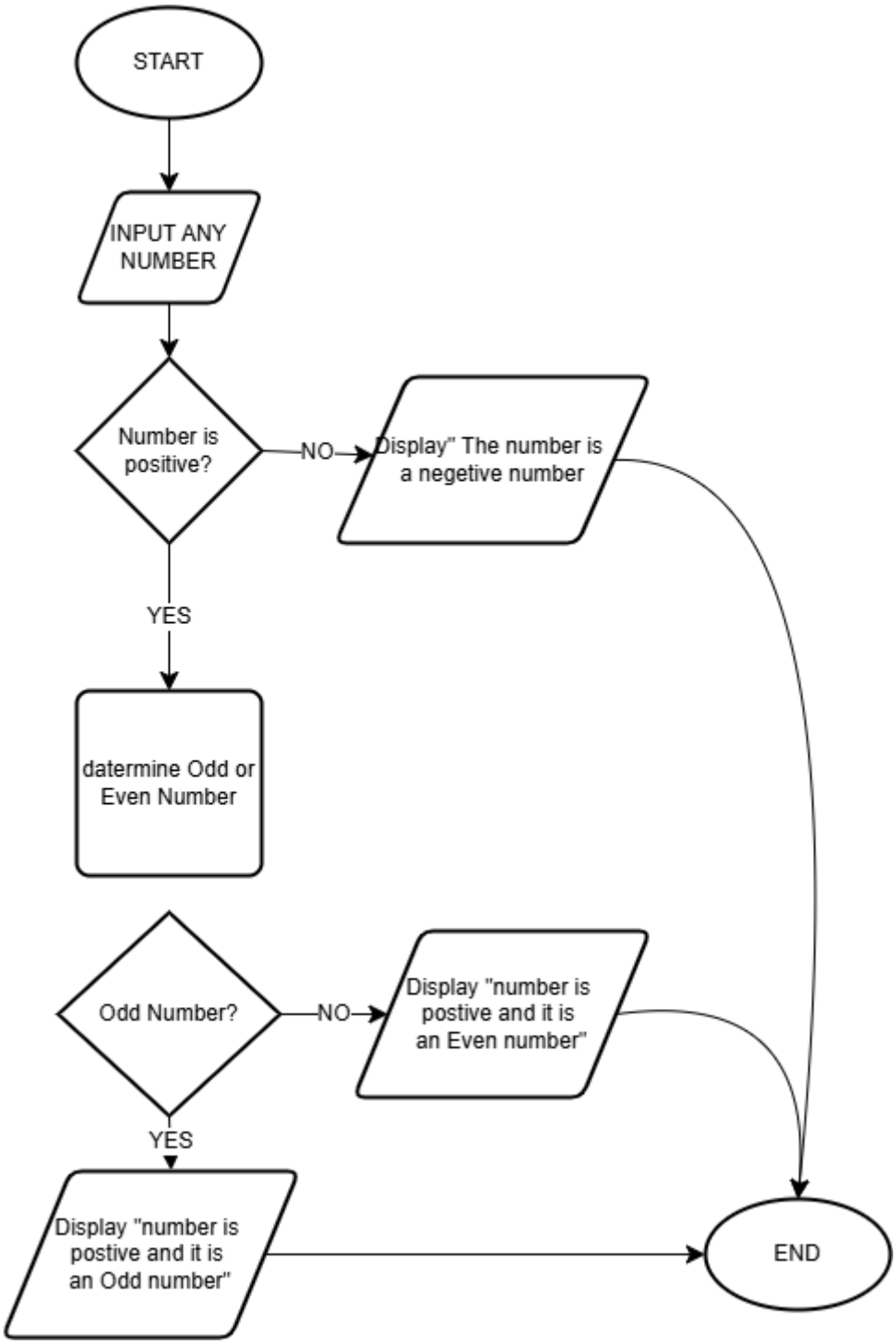
1.1 Create an IPO Analysis. **(9 marks)**

Answer:

INPUT	**Any number positive or negative**
PROCESS	**Check if the number is positive, determine if it's odd or even.**
OUTPUT	**IF positive, Display "**Positive number is Odd/Even Number**"** **IF negative, Display** The number is a Negative Number!"

1.2 Create flowchart that reflects the answer in 1.1. **(10 marks)**

Answer:



1.3 Create a C++ program based on the answers in 1.1 and 1.2. Attach the Screenshot of the output. **(21 marks)**

Answer:

(Total: 40 marks)

END OF PRACTICAL ASSESSMENT ANSWER PAPER